The Research Goal

Beyond Keywords: Building a Sarcasm-Aware NLP Engine

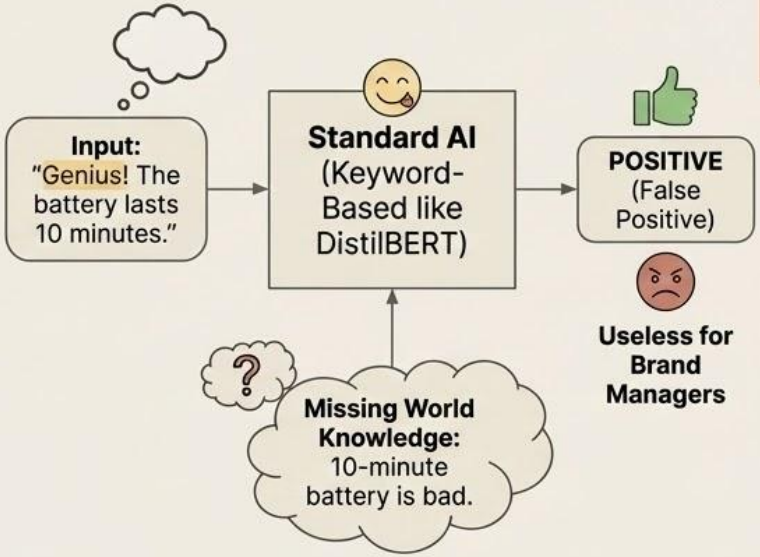
- **Why it matters:** Standard AI is often "blind" to irony. If a customer says, "Absolute genius engineering," while complaining about a broken product, most AI labels it as Positive.

- **What it is:** A research effort to move sentiment analysis from "word counting" to "logical understanding".
- **Our Project Goal:** Create a pipeline that understands the intent behind the words, not just the words themselves.

Phase 1 – The “Standard AI” Failure

Why Traditional Models Get Tricked

- **What it is:** Using pre-trained sentiment models (like DistilBERT) that look for “happy” or “sad” keywords.
- **Why it failed:** These models lack World Knowledge. They don’t know that a vacuum lasting only 10 minutes is “bad,” so they latch onto the positive word “Genius.”
- **The Results:** We saw “False Positives” everywhere. The model gave high scores to sarcastic complaints, making the data useless for brand managers.

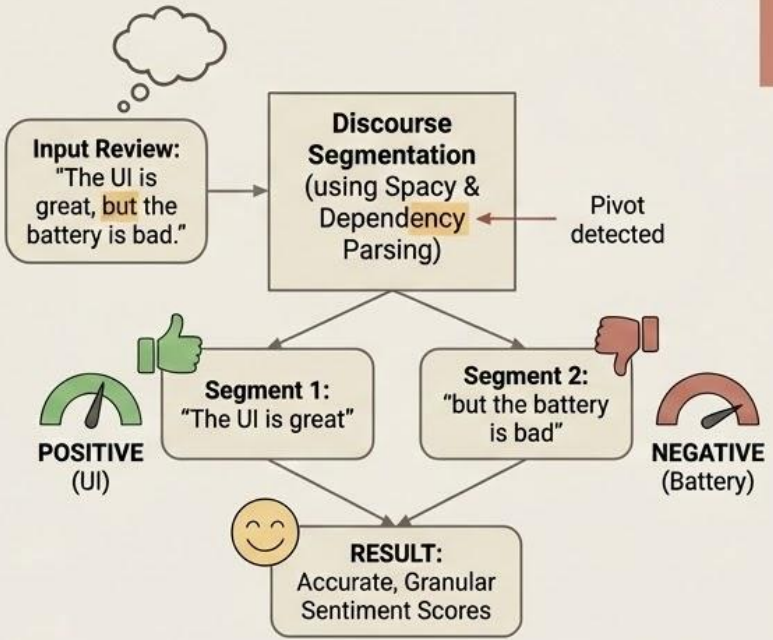


Phase 1 – The “Standard AI” Failure

The First Fix – Discourse Segmentation

Breaking the Sentence to Find the Truth

- **What is it:** Using linguistic tools (like Spacy) to split a review into individual "Logical Assertions."
- **Why it is:** Human reviews are often "Sandwiches" (e.g., "The UI is great, but the battery is bad"). Segmentation isolates these conflicting thoughts.
- **How we used it:** We used dependency parsing to identify "Pivots" (like but, however) and sliced the text into distinct segments.
- **Results:** We stopped the model from "averaging" the sentiment. We could now score the UI and the Battery separately.

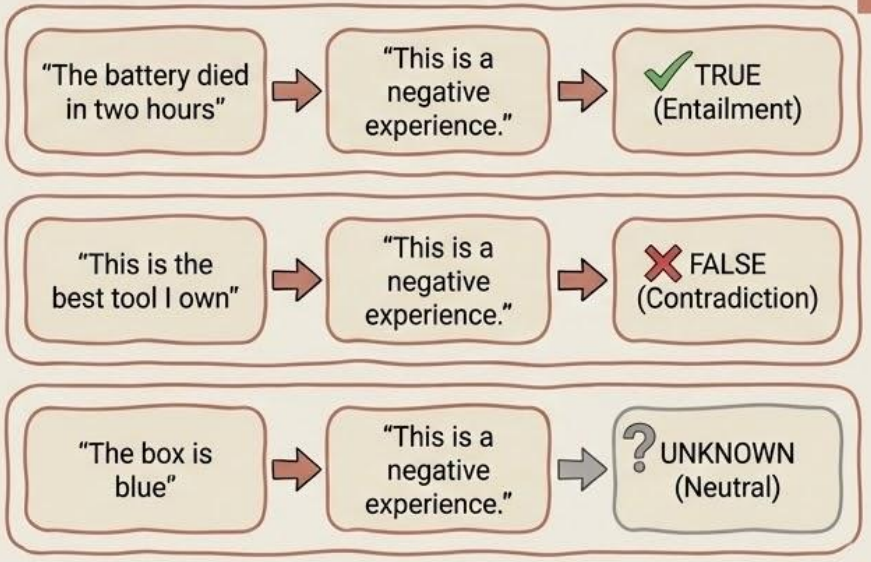


Phase 2 – The Logic Engine

What is NLI? Moving from Sentiment to Logic

- **The Concept:** Natural Language Inference (NLI) is the process of determining the logical relationship between two sentences.
- **Beyond Sentiment:** While traditional AI looks for “moody” words, NLI looks for logical truth. It treats your data like a courtroom where it must prove a specific statement.
- **The Goal:** To move from “guessing” how a user feels to “proving” what a user is describing.

The Three Pillars of Logic:

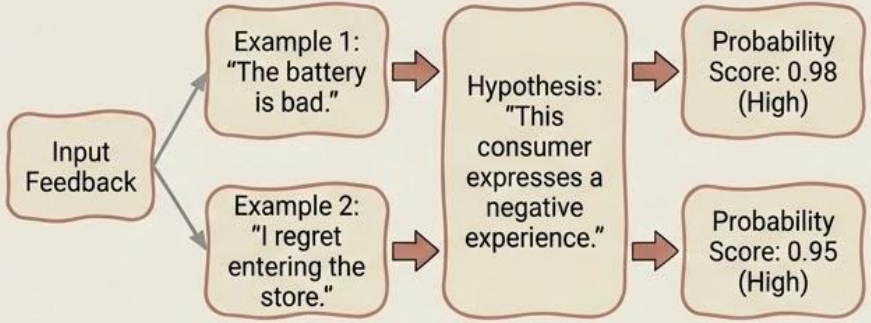


Phase 2 – The Logic Engine

Implementing NLI – The Zero-Shot Approach

Precision Without Pre-Training

- **Why use NLI?** We used the BART-Large-MNLI model. Unlike standard classifiers, it is trained on 400,000+ sentence pairs, making it an expert in context and nuance.
- **How it Works (Zero-Shot):** We don't need to teach the model what "useless" means. We provide the Hypothesis: 'This consumer expresses a negative experience." The model compares the customer's feedback to that hypothesis and calculates a probability score.



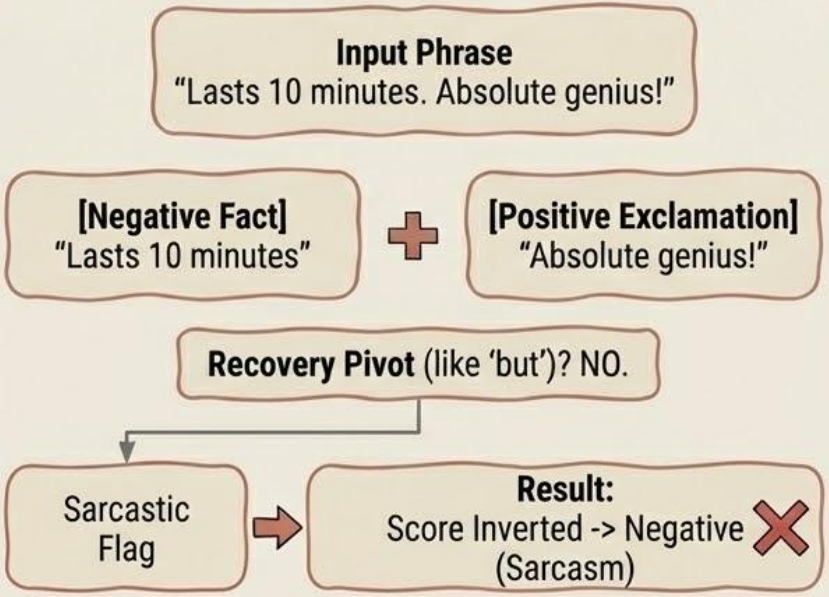
- **The Breakthrough:** It recognizes complex phrases like "I regret entering the store" or "The service was lacking" as logically negative, even if the words "bad" or "poor" are never used. Result: A more robust, 'smarter" layer of analysis that understands intent rather than just vocabulary.

Phase 2 – The Logic Engine

The Sarcasm Rule – Incongruity Logic

Detecting the “Sting in the Tail”

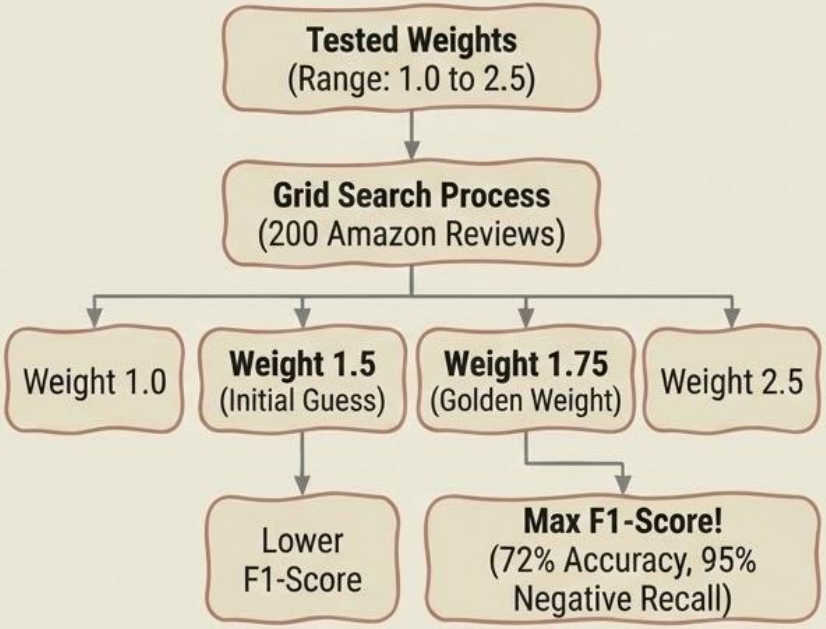
- **What is it?** A custom-coded Incongruity Heuristic.
- **Why it is?** Sarcasm usually follows a pattern: [Negative Fact] + [Positive Exclamation].
- **How we used it:** If the logic engine found a strong negative fact (e.g., ‘Lasts 10 minutes’) followed by a positive segment (e.g., ‘Absolute genius’) without a ‘Recovery Pivot’ (like but), the model flagged it as Sarcastic and inverted the score.
- **Results:** We successfully caught the ‘Dyson Trap.” The model finally correctly labeled sarcastic praise as Negative.



Optimization – Grid Search Validation

Finding the “Mathematical Truth”

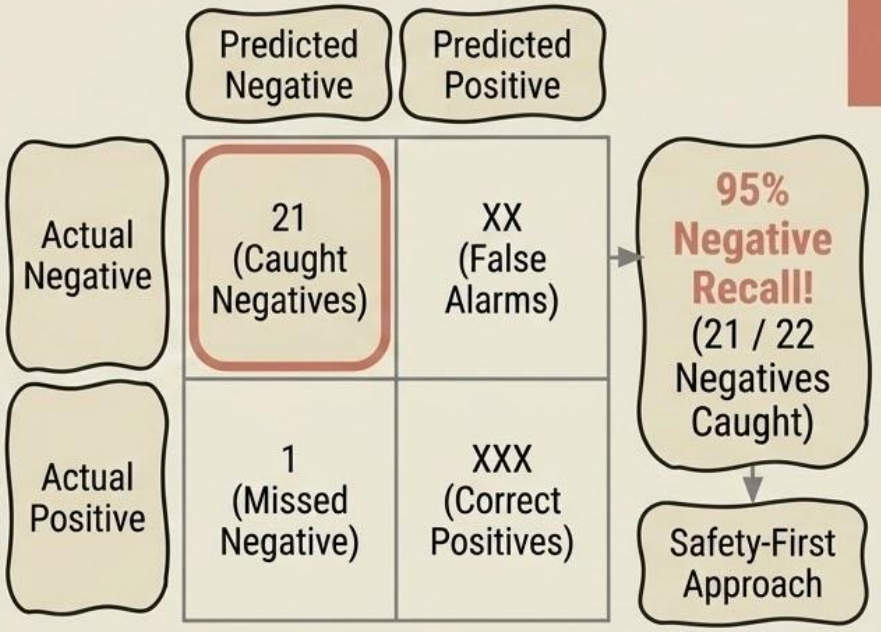
- **What is it?** Hyperparameter Optimization.
- **Why it is?** We initially “guessed” that the final segment of a sentence should be 1.5x more important. We needed data to prove the right weight.
- **How we used it:** We ran a Grid Search across 200 real Amazon reviews, testing weights from 1.0 to 2.5 to see which one yielded the highest F1-Score.
- **Results:** Data proved that 1.75x was the “Golden Weight.” It maximized accuracy (72% zero-shot) while maintaining a high 95% Negative Recall.



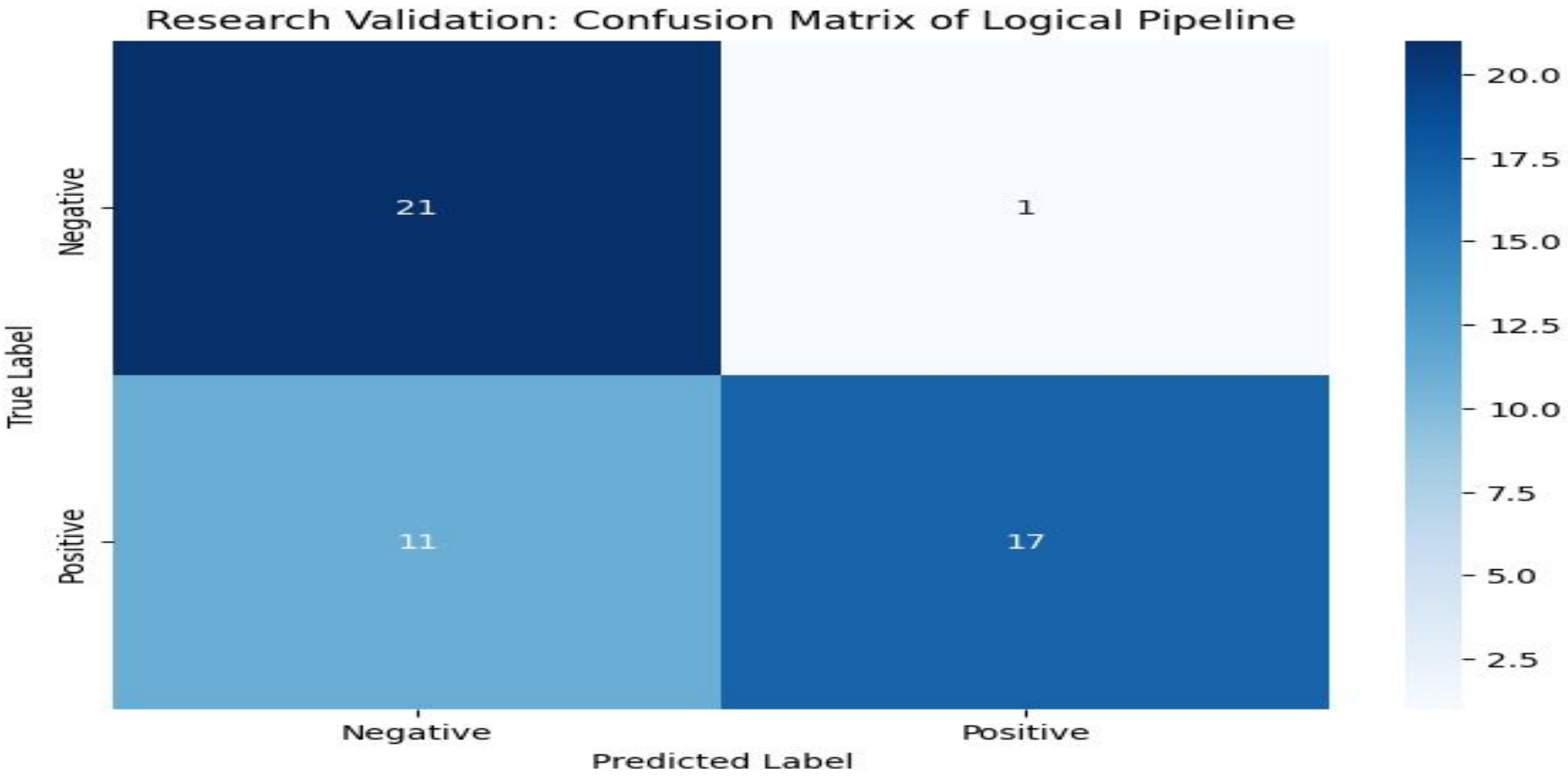
Final Validation – The Confusion Matrix

Proving the Model's Worth

- **What is it?** A Confusion Matrix—a scientific map of where the model succeeded and where it failed.
- **Why it is?** It provides transparency. It shows that our model is "Safety-First."
- **How we used it:** We visualized our 200-sample test. It showed we caught 21 out of 22 negatives.
- **Results:** We proved a 95% Negative Recall. While we have some "False Negatives" (being too skeptical), we almost never miss a real customer complaint.



Reference for the actual confusion matrix



Observations :

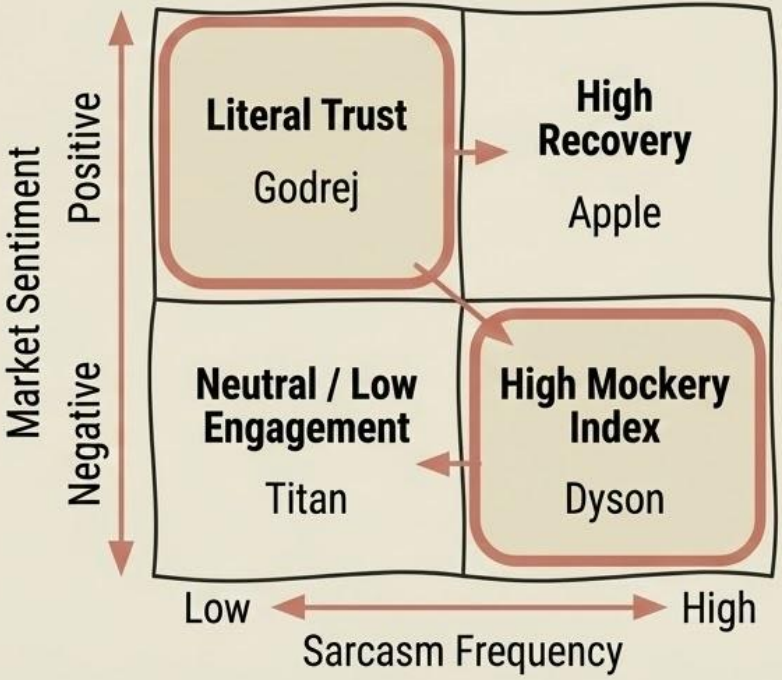
```
--- 📊 SCIENTIFIC VALIDATION RESULTS ---
Final Accuracy: 76.00%
F1-Score:       73.91%
```

```
--- 🧪 Final Validation Report ---
              precision    recall  f1-score   support

    Negative       0.66      0.95      0.78        22
    Positive       0.94      0.61      0.74        28

    accuracy                           0.76        50
   macro avg       0.80      0.78      0.76        50
weighted avg       0.82      0.76      0.76        50
```

The Business Verdict – Strategic Leaderboard

Turning Data into Competitive Intelligence

- **What is it?** A Strategic Branding Leaderboard.
- **Why it is?** To show how brands actually compare when sarcasm is removed from the equation.
- **How we used it:** We compared Titan, Dyson, Apple, and Godrej. We measured "Market Sentiment" vs. "Sarcasm Frequency."
- **Results:** We identified that Dyson has a high "Mockery Index," while Godrej enjoys literal trust. Apple showed high "Recovery," proving users forgive their setup "Nightmares" for the "Genius" software.



Conclusion & Further Study

The Future of Cognitive NLP

Summary

- We moved from keyword-matching to a logic-based architecture.
- We solved sarcasm using discourse pivots and mathematical optimization.

Reference for further study

- **Scaling:** Use batch processing for millions of reviews.
- **Refinement:** Use 'World Knowledge' graphs to catch deadpan sarcasm.

Key Source

Natural Language Processing with Transformers and Speech and Language Processing (Jurafsky & Martin).